

# Provenance.ai — The AI Trust Layer for High-Value Asset Authentication & Intelligence

## The Problem That's About to Explode

By 2026, AI-generated forgeries have become indistinguishable from authentic works. A single Midjourney-generated “Basquiat” sold for \$2.1M before detection. The FBI reports a 340% increase in luxury goods counterfeiting using generative AI. The \$68B art market, \$350B luxury goods market, and \$80B collectibles market face an existential trust crisis.

Traditional authentication relies on: - Human experts (scarce, expensive, inconsistent) - Paper certificates (easily forged) - Siloed databases (fragmented, incomplete) - Reactive detection (too late)

## The Provenance.ai Solution

An end-to-end AI-powered trust infrastructure that: 1. **Authenticates** assets using multimodal AI analysis 2. **Tracks** provenance on immutable blockchain ledger 3. **Predicts** value with market intelligence 4. **Enables** secure transactions and fractional ownership

---

## Market Opportunity

**Total Addressable Market (TAM): \$520B**

| Segment        | Market Size | Authentication Need                      |
|----------------|-------------|------------------------------------------|
| Fine Art       | \$68B       | Forgery rates 20%+ for pre-1960 works    |
| Luxury Goods   | \$350B      | \$500B counterfeits annually             |
| Collectibles   | \$80B       | Sports cards, memorabilia fraud epidemic |
| Wine & Spirits | \$15B       | 20% of fine wine is counterfeit          |
| Classic Cars   | \$10B       | VIN fraud, odometer tampering            |

**Serviceable Addressable Market (SAM): \$45B**

Transactions requiring premium authentication and provenance services.

**Serviceable Obtainable Market (SOM): \$2.5B by Year 5**

Capturing 5.5% of premium authentication market.

---

## Technology Deep Dive

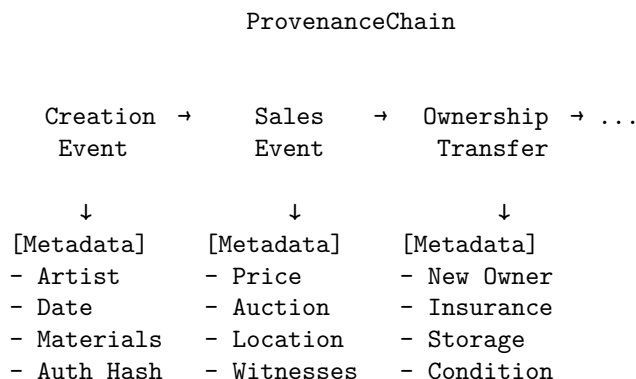
### 1. Multi-Modal Authentication Engine

**Visual Analysis Stack:** - Canvas age estimation via craquelure pattern analysis - Brushstroke kinematics modeling (unique to each artist) - Pigment composition spectral analysis - Frame and stretcher dating - Signature dynamics verification

**Material Science Integration:** - XRF (X-ray fluorescence) data ingestion - Infrared reflectography analysis - UV fluorescence patterns - Carbon-14 dating correlation - Mass spectrometry integration

**Provenance Document AI:** - OCR and handwriting analysis of historical documents - Cross-reference with gallery and auction records - Exhibition history verification - Ownership chain reconstruction - Red flag detection (gaps, inconsistencies)

## 2. ProvenanceChain™ Blockchain Layer



**Key Features:** - Zero-knowledge proofs for private transactions - Multi-party verification consensus - Cross-chain interoperability - GDPR-compliant data handling - Immutable but with privacy layers

## 3. Market Intelligence Platform

**Predictive Valuation:** - Real-time auction monitoring across 500+ houses - Private sale data aggregation - Artist career trajectory modeling - Macro-economic correlation - Taste trend prediction

**Risk Scoring:** - Theft database integration (FBI, INTERPOL, Art Loss Register) - Sanctions screening (OFAC compliance) - Anti-money laundering (AML) analysis - Beneficial ownership identification - Repatriation risk assessment

---

## Product Suite

### For Collectors & Investors

**Provenance Vault** — Personal Collection Management - Digital twin of every owned asset - Real-time valuation dashboard - Insurance documentation - Inheritance planning tools - Mobile authentication scanner

**Price:** \$199/month or 0.5% AUM

### For Auction Houses & Dealers

**Provenance Pro** — Institutional Authentication Suite - Bulk authentication workflows - API integration with existing systems - White-label reports - Expert marketplace access - Compliance documentation

**Price:** \$2,500/month + per-transaction fees

### For Insurance & Financial Services

**Provenance Enterprise** — Risk & Valuation Infrastructure - Underwriting decision support - Claims verification - Collateral assessment - Portfolio risk analysis - Regulatory reporting

**Price:** Custom enterprise contracts (\$500K-\$5M annually)

### For Regulatory & Law Enforcement

**Provenance Shield** — Investigation & Compliance Platform - Stolen property tracking - Sanctions screening - Money laundering detection - Expert witness support - International cooperation tools

**Price:** Government contracts

---

## Competitive Landscape

| Competitor                 | Approach            | Limitation                    |
|----------------------------|---------------------|-------------------------------|
| Artory                     | Blockchain registry | No AI authentication          |
| Verisart                   | NFT certificates    | Limited to digital provenance |
| The Art Genome Project     | Art discovery       | No authentication focus       |
| Traditional Authentication | Human experts       | Expensive, slow, inconsistent |

**Provenance.ai Moat:** 1. **Data Network Effects** — Each authentication improves our models 2. **Institutional Partnerships** — Exclusive access to historical records 3. **Regulatory Capture** — Becoming the standard for AML compliance 4. **Multi-Modal Fusion** — Only platform combining AI + blockchain + material science

---

## Go-to-Market Strategy

### Phase 1: Credibility (Months 1-12)

#### “Win the Experts”

- Partner with 3-5 prestigious museums for training data
- Acquire authentication firm with deep expertise
- Publish breakthrough detection research
- Build advisory board of renowned experts
- Authenticate 10,000 works pro-bono for credibility

**Key Metrics:** - 5 museum partnerships - 50,000 authenticated works - 3 peer-reviewed publications

### Phase 2: Market Entry (Months 13-24)

#### “Anchor the Market”

- Launch with major auction house (Christie’s, Sotheby’s target)
- Insurance carrier partnerships (AXA XL, Hiscox)
- Luxury brand anti-counterfeiting deals
- Free tier for individual collectors
- PR campaign around major forgery detection

**Key Metrics:** - \$10M ARR - 2 top-5 auction house deals - 100,000 active users

### Phase 3: Scale (Months 25-48)

#### “Become the Standard”

- API-first platform for ecosystem integration
- Fractional ownership marketplace launch
- Regulatory compliance certification
- International expansion (EU, Asia)
- M&A of regional authentication companies

**Key Metrics:** - \$100M ARR - 1M authenticated assets - 50% market share in premium segment

---

## Business Model

### Revenue Streams

| Stream                  | Year 1      | Year 3        | Year 5        |
|-------------------------|-------------|---------------|---------------|
| SaaS Subscriptions      | \$2M        | \$25M         | \$150M        |
| Per-Authentication Fees | \$3M        | \$40M         | \$200M        |
| Enterprise Contracts    | \$1M        | \$30M         | \$300M        |
| Data & Intelligence     | \$0.5M      | \$15M         | \$100M        |
| Transaction Fees        | \$0.5M      | \$20M         | \$150M        |
| <b>Total</b>            | <b>\$7M</b> | <b>\$130M</b> | <b>\$900M</b> |

### Unit Economics

**Per-Authentication:** - Average fee: \$150 - COGS (compute, experts): \$25 - Gross margin: 83%

**Enterprise Contract:** - Average ACV: \$750K - Implementation cost: \$75K - Gross margin: 90%

### Path to Profitability

| Metric        | Year 1 | Year 3 | Year 5 |
|---------------|--------|--------|--------|
| Revenue       | \$7M   | \$130M | \$900M |
| Gross Profit  | \$5M   | \$110M | \$765M |
| OpEx          | \$15M  | \$80M  | \$400M |
| EBITDA        | -\$10M | \$30M  | \$365M |
| EBITDA Margin | -143%  | 23%    | 41%    |

## Technology Roadmap

### Year 1: Foundation

- Multi-modal authentication engine v1
- ProvenanceChain blockchain launch
- Mobile authentication app
- API platform beta
- Core integrations (auction houses)

### Year 2: Intelligence

- Predictive valuation engine
- Risk scoring platform
- Automated document analysis
- Real-time market monitoring
- Expert marketplace

### Year 3: Ecosystem

- Fractional ownership infrastructure
- Cross-chain interoperability
- Advanced AML/KYC integration
- IoT sensor integration (smart frames)
- AR/VR collection visualization

## Year 4+: Dominance

- Autonomous authentication agents
  - Global regulatory standard certification
  - Quantum-resistant cryptography
  - Neural authentication interfaces
  - Physical AI inspection robotics
- 

## Team Requirements

### Founding Team

**CEO** — Art market veteran + tech background - Former auction house executive or gallery owner - Deep collector relationships - Fundraising experience

**CTO** — AI/ML + blockchain expertise - PhD-level computer vision background - Blockchain protocol experience - Scaled AI systems before

**Chief Scientist** — Art authentication expert - Museum conservation background - Art history PhD - Industry credibility

**Chief Commercial Officer** — Enterprise sales - Luxury goods or financial services - Global sales experience - Partnership development

### Key Hires (Year 1)

- Head of AI/ML (computer vision focus)
- Head of Blockchain Engineering
- Head of Business Development
- Head of Compliance
- 3-5 authentication experts
- 10 engineers

### Advisory Board

- Museum directors (Met, Louvre, British Museum)
  - Auction house veterans
  - Art law attorneys
  - Insurance executives
  - FBI Art Crime Team alum
- 

## Funding Strategy

### Seed Round: \$8M

**Use of Funds:** - Founding team (40%) - Technology development (35%) - Museum partnerships (15%) - Operations (10%)

**Target Investors:** - Art-focused family offices - Luxury-tech crossover VCs - Strategic angels (auction houses)

### Series A: \$30M (Month 18)

**Use of Funds:** - Scale engineering team (35%) - Enterprise sales (30%) - Marketing & partnerships (20%) - International expansion (15%)

**Target Investors:** - Andreessen Horowitz (fintech/crypto) - Benchmark - LVMH Ventures - Sotheby's Financial Services

**Series B: \$100M (Month 36)**

**Use of Funds:** - Global expansion (40%) - M&A (30%) - R&D (20%) - Working capital (10%)

## Risk Analysis & Mitigation

### Technology Risks

| Risk                       | Probability | Impact   | Mitigation                                         |
|----------------------------|-------------|----------|----------------------------------------------------|
| AI detection defeated      | Medium      | High     | Continuous model improvement, multi-modal approach |
| Blockchain security breach | Low         | Critical | Multi-chain redundancy, insurance                  |
| Expert consensus failures  | Medium      | Medium   | Distributed expert network, appeals process        |

### Market Risks

| Risk                 | Probability | Impact | Mitigation                     |
|----------------------|-------------|--------|--------------------------------|
| Market downturn      | Medium      | Medium | Diversify across asset classes |
| Competitor emergence | High        | Medium | Speed to market, partnerships  |
| Regulatory changes   | Low         | Medium | Proactive compliance, lobbying |

### Operational Risks

| Risk                                 | Probability | Impact   | Mitigation                             |
|--------------------------------------|-------------|----------|----------------------------------------|
| Key expert departure                 | Medium      | High     | Equity incentives, knowledge capture   |
| Partnership failure                  | Medium      | Medium   | Diversified partnerships               |
| Reputational damage (false positive) | Low         | Critical | Conservative authentication, insurance |

## Success Metrics & Milestones

### Year 1

- ☐ 50,000 assets authenticated
- ☐ 5 museum partnerships signed
- ☐ 2 auction house pilots launched
- ☐ \$7M revenue
- ☐ Series A closed

## Year 2

- ☐ 500,000 assets authenticated
- ☐ Market leader in AI authentication
- ☐ 10 enterprise customers
- ☐ \$50M revenue
- ☐ International presence (EU)

## Year 3

- ☐ 2M assets authenticated
- ☐ Industry standard for provenance
- ☐ 50% premium market share
- ☐ \$130M revenue
- ☐ Path to profitability clear

## Year 5

- ☐ 10M assets authenticated
  - ☐ \$900M revenue
  - ☐ \$5B+ valuation
  - ☐ IPO ready
- 

## Why Now?

1. **AI Forgery Crisis** — Generative AI has made counterfeiting trivially easy. The problem is exploding NOW.
  2. **Blockchain Maturity** — Enterprise blockchain infrastructure is finally production-ready.
  3. **Regulatory Pressure** — EU and US tightening AML requirements for art transactions.
  4. **Market Liquidity** — High-value assets increasingly financialized (collateral, fractional ownership).
  5. **Institutional Appetite** — Art funds, family offices demanding institutional-grade infrastructure.
  6. **Technology Convergence** — Computer vision, materials science, and distributed ledger tech have all matured simultaneously.
- 

## The \$10B Exit Path

### Potential Acquirers:

| Company              | Strategic Rationale                      | Estimated Value |
|----------------------|------------------------------------------|-----------------|
| LVMH                 | Luxury authentication + brand protection | \$8-15B         |
| Mastercard/Visa      | High-value transaction infrastructure    | \$10-20B        |
| Sotheby's/Christie's | Vertical integration                     | \$5-10B         |
| Bloomberg            | Alternative asset data                   | \$8-12B         |
| Insurance Giants     | Risk assessment infrastructure           | \$6-10B         |

**IPO Path:** - 2030 target at \$15B+ valuation - Comparable: Datadog, MongoDB (data infrastructure) - 40%+ gross margins, 30%+ growth

---

## Conclusion

Provenance.ai isn't just building a product — it's creating the trust infrastructure for a \$500B market facing an existential crisis.

As AI makes deception trivial, authenticity becomes priceless.

We're building the standard.

---

*“In a world where anything can be faked, proof of authenticity becomes the most valuable asset of all.”*

**Contact:** founders@provenance.ai

**Deck:** Available upon request

---

© 2026 Provenance.ai — All Rights Reserved